From Pairwise Alignment to Cluster Stability: A Paired Cross-Lingual Study of English and Vietnamese App Descriptions

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Abstract. We compare vector correspondence, partition agreement and taxonomy alignment in an observational study of bilingual Google Play descriptions. RQ1 pairs English-source and Google-translated Vietnamese descriptions for 13,657 apps; RQ2 compares eight representation recipes in a mixed-provenance Vietnamese benchmark of 23,883 apps and in provenance-specific clustering refits. Same-app pairs are closer than the anchor-specific nonmatching median on average in every paired condition, but cross-language partition agreement is lower than initialization-only within-language agreement. The descriptive difference S ranges from -0.155594 to -0.053489, while the direction of English-versus-Vietnamese taxonomy alignment varies by recipe. A native lexical changed-text control on 13,230 eligible apps yields cross-partition adjusted mutual information (AMI) of 0.487984 between English halves versus 0.681595 between full English and Vietnamese views, showing that a larger deficit can arise without a language change. E5, BGE and Qwen occupy the highest three observed taxonomy-alignment means in both recorded provenance strata and input conditions, with changing internal order. A bootstrap multiplicity-sensitivity diagnostic identifies weighting-sensitive centering in historical fixed-label AMI resampling; it supplies no corrected confidence interval. The findings characterize the tested catalogue and recipes without identifying a causal translation effect.

Keywords: Cross-lingual text embeddings; English–Vietnamese; App descriptions; Clustering stability; Adjusted mutual information; Machine translation.

1. Introduction

App descriptions support the comparison and organization of mobile applications. Prior work has evaluated text-based app similarity [1], Android app categorization [2] and classification using word embeddings [7]. For a bilingual catalogue, similarity between paired descriptions alone does not reveal whether the language views induce comparable groups or align with the same taxonomy. This study examines those outcomes together on the same app identities.

Multilingual sentence-embedding methods support comparison across languages [3, 4], while embedding benchmarks evaluate multiple task types [5, 6]. A paired app catalogue provides a common unit for comparing vectors, independently fitted partitions and taxonomy alignment: each English and Vietnamese observation refers to the same app. Translation artifacts [9] and language-dependent tokenization [11] motivate explicit reporting of text provenance and input handling.

We evaluate English-source and Google-translated Vietnamese Google Play descriptions at three levels. Vector correspondence compares each paired description with an anchor-specific nonmatching reference. Partition agreement compares cross-language groupings with initialization-only agreement within each language. Taxonomy alignment measures agreement between those groupings and store genres. Evaluating the three outcomes on the same apps permits a joint descriptive interpretation without assuming that success on one guarantees success on another.

The paired analysis contains 13,657 apps. A complementary benchmark contains 23,883 Vietnamese descriptions with mixed recorded translation provenance. We ask:

- **RQ1:** How do English-source and Google-translated Vietnamese descriptions of the same apps correspond at the vector and partition levels, and how do their partitions align with the store taxonomy?
- **RQ2:** How do the evaluated representation and input recipes align with the store taxonomy in the mixed-provenance Vietnamese benchmark and within its recorded provenance strata?

The contribution is an empirical account of positive same-app correspondence alongside an initialization-relative partition deficit and recipe-dependent taxonomy contrasts. A same-language lexical control shows that a larger deficit can arise without a language change, limiting a translation-specific interpretation. Provenance-specific refits establish that the leading observed recipe pattern is present in both recorded strata. An uncertainty sensitivity diagnostic qualifies the historical AMI app-resampling evidence. These findings characterize the tested catalogue and complete recipes; they do not identify a causal translation effect or a mechanism linking vector proximity to partition changes.

2. Related work

2.1. App descriptions and embedding evaluation

App-description studies already compare clustering solutions and their relation to store categories. Al-Subaihin et al. evaluate description-based similarity techniques and the partitions they produce [1]; Alecci et al. revisit Android app categorization [2], and Ebrahimi et al. study classification using word embeddings [7]. Our focus is the paired English-source and Google-translated Vietnamese views of the same apps, with within-language initialization agreement as the reference for partition comparison. Vietnamese conversation clustering [8] provides a related application of clustering in a different text domain.

Multilingual embedding research includes knowledge distillation from monolingual sentence embeddings [3] and language-agnostic bilingual sentence alignment [4]. The Massive Text Embedding Benchmark (MTEB) [5] and its Vietnamese counterpart, VN-MTEB [6], evaluate embeddings across task collections. The present study adds a domain-specific paired comparison of correspondence, partition agreement and store-taxonomy alignment on common app identities. Translation artifacts in cross-lingual transfer [9] motivate interpreting the bilingual contrast within its recorded translation conditions.

2.2. Representation and input recipes

PhoBERT [13], SimCSE [14] and DeBERTaV3 [15] provide pretraining and sentence-embedding backgrounds for the evaluated Vietnamese checkpoints. Their exact inference configurations, including segmentation and pooling, are specified in Tables A1 and A2.

The multilingual recipes use E5 [16], BGE-M3 [17] and Qwen3 Embedding [18]. The lexical baseline combines term weighting [19] and latent semantic projection [20]. We compare the complete inference recipes, whose implementation is detailed in Section 3.2, rather than attributing their differences to architecture alone.

Language-dependent tokenization [11] and long-text clustering studies [12] motivate explicit input-handling rules. Section 3.3 distinguishes the shared character-prefix condition from each recipe's native token cap.

2.3. Partition agreement and representation geometry

Chance-adjusted mutual information provides a basis for comparing partitions and external labels [22]. Clustering-stability research emphasizes that the perturbation and algorithm define the stability question [31]. We accordingly distinguish initialization-only agreement from agreement after changing text and language, and use S as a descriptive difference between those references. The subtraction describes a reference-dependent comparison rather than a general stability measure.

Contextual-representation geometry can also affect how similarity is interpreted. Ethayarajh studies anisotropy in contextual representations [30], while Su et al. propose whitening sentence representations [32]. These studies motivate attention to geometry when interpreting recipe-dependent similarity. This study evaluates the unwhitened representations.

3. Data and methods

3.1. Study design and cohorts

The observational design links vector correspondence, partition agreement and taxonomy alignment through app identity. RQ1 uses paired English-source and Google-translated Vietnamese descriptions; RQ2 evaluates the larger mixed-provenance Vietnamese corpus and its recorded strata. Both use the same store-taxonomy reference.

We collected 24,756 Google Play listings on 2026-09-17 using `google-play-scraper`. Successive Vietnamese selection stages retain 24,454 descriptions with a Vietnamese-specific character, 24,285 with the Vietnamese eligibility flag, and 24,256 with at least 50 characters. The eligibility flag requires Vietnamese-specific characters to comprise at least 0.1 of alphabetic characters. Near-deduplication yields 23,883 unique apps. MinHash locality-sensitive hashing (LSH) uses lowercase word trigrams, 128 permutations and an estimated Jaccard threshold of 0.8 to form connected duplicate components. Representatives are selected deterministically by descending install counts and ascending app identifier. The Vietnamese input is the cleaned description field.

Recorded translation provenance divides this benchmark into 16,394 translation-flagged apps, 7,466 explicitly not-flagged apps and 23 with unknown status. A false translation flag does not verify human authorship, native fluency or the absence of unrecorded translation. We therefore retain the term *not-flagged* throughout.

For the paired study, 16,375 of the flagged apps have both cleaned descriptions of at least 50 characters. Agreement between fastText and py3langid identifies 13,723 English-source candidates; English-side near-deduplication leaves 13,657 pairs. The English input is the cleaned original description. English-source denotes detector consensus on this selected text, and Google-translated Vietnamese

denotes the recorded translation condition. Neither designation constitutes an independent assessment of language or translation quality.

Table 1. Dataset overview. All five experimental cohorts contain the 49 store genre categories. Provenance strata are used for independent clustering refits; the English-half control is restricted to eligible pairs. Minimum and maximum apps per genre are reported in Table B3.

Analysis	Apps	Genre categories represented
Paired EN–VI study	13,657	49
Mixed-provenance Vietnamese benchmark	23,883	49
Same-language changed-text control	13,230	49
Translation-flagged refits	16,394	49
Not-flagged refits	7,466	49

The reference categories are the 49 distinct store genre identifiers, rather than display names that may be duplicated. They form a flat, single-label commercial taxonomy, not semantic ground truth. The cohort has no verified probability-sampling frame. Median cleaned length is 1,874 Unicode characters in the Vietnamese benchmark and 1,670 English versus 1,888 Vietnamese characters in the paired cohort. The mean paired Vietnamese-to-English character ratio is 1.129739. These length differences motivate explicit input conditions rather than an assumption that paired descriptions expose equal amounts of text.

3.2. Representation recipes

We compare eight complete representation recipes in the Vietnamese benchmark and the lexical, E5, BGE and Qwen recipes in the paired study. Each recipe includes its checkpoint, tokenizer, segmentation, pooling, prefix and input cap (Tables A1 and A2). TF-IDF–LSA combines term weighting [19] with latent semantic projection [20]; the neural backgrounds are described in Section 2.2. The comparisons characterize these recipes, without isolating architecture effects.

The lexical recipe lowercases text and uses regex word tokens with at least two word characters, unigrams and bigrams, a minimum document frequency of 5, a maximum document-frequency fraction of 0.5, sublinear term frequency and L2-normalized TF-IDF. Truncated SVD retains 300 dimensions with random state 42, followed by L2 normalization. English and Vietnamese paired inputs are fitted jointly to one vocabulary and SVD map within each input regime. This is a transductive catalogue representation, with no supervised genre training. The provenance-specific refits retain the globally fitted Vietnamese lexical map.

Sup-SimCSE uses the checkpoint's pooler output. E5 uses the literal prefix "query: ", including its trailing space. BGE uses only its dense CLS representation, and Qwen receives no instruction. No centering or whitening is applied.

3.3. Input conditions

The Vietnamese benchmark compares a 950-character common-content condition with native input. The common-content condition takes at most 950 cleaned Unicode characters, moving the end to the preceding whitespace boundary. If no usable boundary exists, or that boundary would leave fewer than 20 characters, it retains the hard prefix. Segmentation where configured, the runtime prefix, tokenization and the recipe's native cap follow this cut. Native input retains the full cleaned text until the recipe's own token cap; the native lexical recipe is unbounded.

The 950-character budget follows a tokenizer audit rather than a performance optimum. Among tested candidates of 450, 700, 950, 1,200 and 1,500 characters, it is the largest whose 95th-percentile token length fits all three 256-token audit limiters: PhoBERT-v2, Sup-SimCSE and BKAI. At 950 characters, their median is 192 tokens and their 95th percentile is 236; 383 of 23,883 descriptions still exceed the cap (1.604%). At 1,200 characters, the 95th percentile reaches 295 tokens and fails the criterion. The budget provides a pragmatic shared prefix while allowing occasional truncation.

The paired study compares native input with a 512-token condition. Neural input is capped at the smaller of 512 and the native maximum, including special tokens. TF-IDF–LSA instead retains and rejoins the first 512 lowercase regex word tokens, a different unit from neural subword tokens. E5 has a cap of 512 in both conditions; its separately stored runs are not assumed to have identical vectors. Full-text E5 truncation affects 5,026 English inputs (36.802%) and 6,770 Vietnamese inputs (49.572%). Tables B1 and B2 report description lengths and condition-specific token-cap truncation. Native versus common-content contrasts jointly vary retained text, tokenization and recipe capacity, so they do not isolate a causal context-length effect.

3.4. Clustering and evaluation

We apply standard Euclidean KMeans to L2-normalized vectors with 49 clusters, 10 initializations per fit and outer seeds 42 through 51. The cluster count matches the number of taxonomy categories. Recorded scikit-learn defaults supply k-means++ initialization [21], Lloyd iterations, a maximum of 300 iterations and tolerance of 0.0001.

Centroids are not constrained to the unit sphere, so this procedure is not spherical KMeans. The secondary hierarchical configuration uses condensed cosine distances, average linkage and a maximum-cluster cut at 49.

For vector correspondence, let e_i and v_i denote the English and Vietnamese vectors for app i. We define

$$\Delta\cos_i = \cos(e_i, v_i) - \text{median}_{j \neq i} \cos(e_i, v_j). \quad (1)$$

We report the mean across apps. The nonmatching catalogue includes apps in the same genre. A positive mean indicates correspondence above the anchor-specific nonmatching median; the statistic is neither a retrieval accuracy measure nor a calibrated null comparison. Cosine geometry and the scale of Δcos depend on the representation recipe.

Partition comparisons use adjusted mutual information (AMI), with arithmetic averaging in its normalization [22]. Taxonomy alignment is the mean AMI between genre identifiers and cluster labels across the ten seeds. Cross-language agreement averages all 100 English-by-Vietnamese seed-pair AMIs. Within-language agreement averages the 45 distinct seed-pair AMIs separately in English and Vietnamese. We define the initialization-relative partition difference as

$$S = A_{\text{cross}} - \frac{A_{\text{EN}} + A_{\text{VI}}}{2}. \quad (2)$$

Here, A_{cross} is cross-language agreement, and A_{EN} and A_{VI} are within-language agreement in English and Vietnamese, respectively.

The within-language reference changes initialization while retaining the text; the cross-language comparison changes text and language as well. Negative S therefore records lower cross-language agreement relative to that reference. Its sign is an empirical result, not a mathematical requirement. The 100 cross-language comparisons share partitions and are correlated.

3.5. Uncertainty and sensitivity to app multiplicity

Existing paired-seed Student-t intervals use ten matched-seed differences and nine degrees of freedom. These marginal, unadjusted intervals condition on the fixed cohort, representations and recorded seed panel. The small-panel approximation assumes approximately independent seed blocks and well-behaved paired differences under the specified initialization mechanism. The intervals do not quantify app, population, seed-population or full-pipeline uncertainty.

The app-resampling uncertainty analysis followed availability of the primary point estimates. The changed-text and provenance-refit protocols were fixed before their respective outcomes were examined.

The historical app-resampling procedure drew apps with replacement 1,000 times, retained duplicate occurrences and evaluated AMI on repeated fixed labels, while the reported full-cohort points used each app once. Bootstrap methodology supplies general background [25], but does not establish the calibration of this implementation. Historical AMI percentile ranges are retained only as archival context for uncertainty scale. The separate historical scalar ranges for Δcos are identified as such; the AMI diagnostic below does not assess their coverage.

The bootstrap multiplicity-sensitivity diagnostic compares the first 200 stored replacement draws with the unique app support of each same draw. Let I_b be an original cohort-length replacement vector, U_b its sorted unique app identities, and T_N the unique full-cohort point. WR denotes the stored repeated-label result on I_b; UNIQUE evaluates the same fixed-label quantity with each selected app counted once on U_b. We report the following diagnostic quantities:

$$\begin{aligned} d_{\text{WR}} &= \text{mean}(\text{WR}) - T_N, \\ d_{\text{UNIQUE}} &= \text{mean}(\text{UNIQUE}) - T_N, \\ D_{\text{MULT}} &= \text{mean}(\text{WR} - \text{UNIQUE}). \end{aligned} \quad (3)$$

Medians, Monte Carlo standard errors and means in the two draw halves assess the simulation precision and stability of the paired difference. UNIQUE is a support-matched diagnostic, not a corrected estimator or interval method. No replacement app confidence interval is reported.

3.6. Same-language changed-text control

The native lexical control asks whether changing content within English can also produce a negative initialization-relative partition difference. Eligibility requires at least 40 whitespace-delimited English words, retaining 13,230 paired apps and excluding 427. Each English description is split deterministically at the floor of half its word count into nonoverlapping first and second halves, each containing at least 20 words. All 49 genres remain represented.

One TF-IDF/SVD map fitted to the full paired English and Vietnamese catalogue transforms full English, full Vietnamese and both English halves. KMeans is independently refitted for each view using the same ten-seed procedure, yielding 40 fits and 380 partition AMIs. Both comparisons use identical eligible app support. The protocol was fixed before outcomes, without selective reruns. Because splitting changes content and length differently from translation, the control tests the specificity of a lexical partition deficit, not a causal language effect or a neural-model result.

3.7. Provenance-specific clustering

For the provenance-stratified analysis, we refit KMeans independently within the translation-flagged and not-flagged strata, retaining the original vectors and global lexical map. The 23 unknown-provenance apps are excluded only from these refits. Eight recipes, two input conditions, two strata and ten seeds produce 320 completed fits. We report each cell's mean taxonomy AMI, sample standard deviation across seeds and descriptive rank, without new app intervals or hypothesis tests. Unequal cohort sizes and genre compositions remain part of the observed comparison.

4. Results

4.1. RQ1: vector correspondence and partition agreement

Same-app pairs are closer than the anchor-specific nonmatching median on average in every evaluated recipe and input condition (Table 2). The differences have recipe-dependent scales, so their magnitudes should not be read as a common ranking of semantic fidelity.

Positive vector correspondence coexists with lower agreement between English and Vietnamese partitions than between initialization variants within either language. Cross-language AMI ranges from 0.657292 to 0.727105, and S is negative in all eight conditions (Table 2). The least negative S is -0.053489 for native BGE; the most negative is -0.155594 for the 512-token lexical condition. The paired observations establish an initialization-relative partition deficit, without isolating its cause.

4.2. RQ1: cross-language taxonomy alignment

The English-minus-Vietnamese taxonomy AMI contrast is positive for BGE and Qwen and negative for E5 and TF-IDF–LSA in both input conditions (Table 3). For native input, it is 0.004849 for BGE, 0.003332 for Qwen, -0.018873 for E5 and -0.015686 for the lexical recipe. The observed point direction therefore depends on recipe. The conditional seed intervals in Table C2 span zero for Qwen in both conditions; its positive point contrasts remain directionally uncertain under the conditional seed analysis. These intervals describe initialization variation, not app or population uncertainty.

4.3. Same-language changed-text control

Changing text within English produces an even larger observed partition deficit in the tested native lexical recipe. On identical support of 13,230 apps, cross-partition AMI is 0.487984 between English halves, compared with 0.681595 between full English and Vietnamese descriptions. The corresponding S values are -0.312124 and -0.121366 (Table 4). Within-view agreements remain near 0.8 in both comparisons, so the larger deficit for English halves accompanies lower agreement across text views rather than uniformly low initialization stability.

The control establishes that a negative initialization-relative difference can arise under same-language content change in the native lexical recipe. English halves discard different content and reduce length differently from translation. The result limits a translation-specific explanation of the lexical deficit; it does not estimate a causal translation effect or extend to neural encoders.

4.4. RQ2: taxonomy alignment in the mixed-provenance Vietnamese benchmark

E5, BGE and Qwen have the highest three observed taxonomy-alignment means under both input conditions in the 23,883-app Vietnamese benchmark (Table 5). E5 leads in the 950-character common-content condition at 0.530464, followed by BGE at 0.524273 and Qwen at 0.518074. Under native input, BGE has the highest mean at 0.527976, closely followed by Qwen at 0.527494, while E5 has a mean of 0.520598. The internal ordering therefore changes with the complete input recipe. Conditional seed intervals for the native-minus-common-content contrasts are reported in Table C3; they do not represent app or population uncertainty.

Native input does not uniformly increase taxonomy alignment. The lexical recipe increases from 0.487014 to 0.498861 and Qwen from 0.518074 to 0.527494, whereas E5 decreases from 0.530464 to 0.520598. PhoBERT changes little in point magnitude. ViDeBERTa has weak alignment in both evaluated conditions, at 0.022101 and 0.027696. These are outcomes of the specified checkpoint and preprocessing/pooling configuration; they do not establish a model-family failure or its cause. Differences in retained content and tokenization prevent attribution of the input contrasts to context length alone.

4.5. Provenance-specific clustering refits

Across the 320 provenance-specific fits, the same three recipes occupy the highest three mean positions within each recorded stratum and input condition. Both strata contain all 49 genres, with minimum genre support of 122 apps in the translation-flagged stratum and 18 in the not-flagged stratum. Table C4 reports all recipes, conditions and strata, including seed standard deviations and descriptive ranks.

With common-content input, both strata order the leading means as E5, BGE and Qwen. Native input changes the translation-flagged order to Qwen,

Table 2. Vector correspondence and partition agreement. Full-cohort results for 13,657 paired apps. Mean Δcos measures same-app correspondence relative to the anchor-specific nonmatching median. Cross-language AMI averages 100 seed-pair comparisons; S subtracts the average within-language agreement from cross-language agreement. Historical scalar ranges for Δcos and the two within-language AMIs are reported in Table C1. No app confidence intervals are attached.

Recipe	Input condition	Mean Δcos	Cross-language AMI	S
BGE-M3	Native	0.457410	0.727105	-0.053489
BGE-M3	512-token	0.457624	0.723512	-0.064320
E5-large	Native	0.110037	0.672716	-0.091737
E5-large	512-token	0.110037	0.672781	-0.091655
Qwen3-Embedding-0.6B	Native	0.602891	0.708592	-0.092793
Qwen3-Embedding-0.6B	512-token	0.599630	0.704336	-0.099476
TF-IDF–LSA	Native	0.064409	0.674614	-0.131111
TF-IDF–LSA	512-token	0.061562	0.657292	-0.155594

Table 3. Cross-language taxonomy alignment. EN and VI are ten-seed mean AMIs against the same store genres for 13,657 paired apps. The contrast is EN minus VI. The existing marginal, unadjusted conditional seed intervals are reported in Table C2; they describe initialization variation, not app or population uncertainty.

Recipe	Input	EN mean	VI mean	EN − VI
BGE-M3	Native	0.524145	0.519295	0.004849
E5-large	Native	0.494553	0.513426	-0.018873
Qwen3-Embedding-0.6B	Native	0.521344	0.518012	0.003332
TF-IDF–LSA	Native	0.469384	0.485070	-0.015686
BGE-M3	512-token	0.524515	0.516705	0.007810
E5-large	512-token	0.495961	0.513426	-0.017466
Qwen3-Embedding-0.6B	512-token	0.519959	0.515427	0.004532
TF-IDF–LSA	512-token	0.469929	0.489281	-0.019352

Table 4. Same-language changed-text control. Both rows use the same 13,230 eligible paired apps and a single shared TF-IDF/SVD map. "First" and "second" refer to the named views. These are refit results on restricted support, separate from the full-cohort results in Tables 2 and 3.

Views	Apps	Cross AMI	Within first	Within second	S
Full EN / full VI	13230	0.681595	0.796252	0.809669	-0.121366
First / second English halves	13230	0.487984	0.802187	0.798028	-0.312124

Table 5. Vietnamese benchmark. Ten-seed mean AMIs for 23,883 apps. The contrast is native minus 950-character common-content. The existing marginal, unadjusted conditional seed intervals are reported in Table C3; they describe initialization variation, not app or population uncertainty.

Recipe	Common-content mean	Native mean	Native – common-content
BGE-M3	0.524273	0.527976	0.003703
BKAI bi-encoder	0.371782	0.384350	0.012568
E5-large	0.530464	0.520598	-0.009865
PhoBERT-v2	0.392279	0.392875	0.000596
Qwen3-Embedding-0.6B	0.518074	0.527494	0.009420
Sup-SimCSE	0.460832	0.466232	0.005401
TF-IDF–LSA	0.487014	0.498861	0.011848
ViDeBERTa	0.022101	0.027696	0.005595

BGE and E5, with means of 0.527655, 0.526832 and 0.520593. The native not-flagged order remains E5, BGE and Qwen, with close means of 0.556867, 0.556296 and 0.555295. The leading three-recipe pattern is therefore not confined to flagged translations in these refits, while the ordering within that group is condition-dependent.

The refits describe the observed cohorts in fixed representation spaces. Unequal stratum sizes, genre compositions and the globally fitted lexical map preclude a causal provenance comparison, and the not-flagged stratum does not verify human authorship.

4.6. Diagnostics and secondary clustering

4.6.1. Bootstrap multiplicity sensitivity

The paired mean WR-minus-UNIQUE difference is positive in all eight S conditions, ranging from 0.001264531 to 0.003401710. Counting each selected app once brings the diagnostic mean closer to the unique full-cohort point in every condition (Table D1). The multiplicity differences are approximately 28.4–40.4% of the corresponding historical S percentile-interval widths. This is a scale comparison, not validation or correction of those intervals.

The paired diagnostic centers are stable across the two halves of the stored draw sequence. Tables D1–D3 report all eight centering comparisons, medians, Monte Carlo precision estimates and split-half summaries; detailed examples appear in Appendix D.

The comparison diagnoses sensitivity to multiplicity weights on fixed partitions. It neither recalibrates historical intervals nor changes the full-cohort points.

4.6.2. Secondary average-linkage clustering

The secondary average-linkage configuration produced concentrated partitions: a single cluster exceeded 90% of apps in 9 of 16 mixed-Vietnamese configurations and 7 of 16 paired-analysis configurations. The paired-analysis configurations count English and Vietnamese separately. This adverse concentration characterizes the tested configuration; it does not establish KMeans superiority. Because the procedures also differ in distance metric and clustering objective, the result cannot be attributed solely to linkage or metric.

5. Discussion

5.1. Correspondence, agreement and taxonomy alignment answer different questions

The paired results show why same-app correspondence alone is insufficient to characterize catalogue organization. Across the tested conditions, positive mean $\Delta\cos$ coexists with lower cross-language partition agreement than initialization-only within-language agreement. Taxonomy alignment adds a separate result: the direction of the English-minus-Vietnamese contrast varies by recipe. A single language ranking cannot summarize these observations.

S summarizes the partition contrast relative to a named perturbation reference: changing text and language versus changing initialization alone. It does not identify why particular apps change groups. The evidence supports a descriptive comparison of the tested representations, rather than a mechanism connecting pairwise geometry to clustering assignments.

5.2. What the changed-text control and provenance refits add

The English-half control shows that the partition deficit depends on the text perturbation. On the same eligible apps, changing content within English yields a larger observed lexical partition deficit than the full English–Vietnamese comparison. Negative S is therefore not specific to a language change in this recipe. Because the halves also shorten descriptions and discard different content, the control cannot determine how much of the bilingual deficit is attributable to language or translation.

The provenance refits show that the leading recipe pattern extends beyond the translation-flagged subset. E5, BGE and Qwen occupy the highest three observed mean positions in both strata and input conditions, while their internal ordering changes. Persistence in the not-flagged stratum extends the descriptive scope of the benchmark finding; it does not establish human-authored Vietnamese performance or a causal provenance effect.

5.3. Recipe and input dependence

Input handling is part of the representation being evaluated. The 950-character condition makes the initial text prefix more comparable across recipes, whereas native input exposes each recipe's own cap. The mixed directions of the native-minus-common-content contrasts show that retaining more text does not uniformly improve observed taxonomy alignment. Differences in tokenization, pooling, segmentation and capacity prevent an architecture-only or causal context-length interpretation.

The low ViDeBERTa taxonomy AMI remains specific to the tested checkpoint and preprocessing/pooling configuration. Anisotropy [30] and whitening [32] provide relevant geometric background, but neither was tested as an explanation or remedy here. The origin of the stored E5 numerical and partition differences also remains unresolved; no controlled evidence assigns them to a software environment.

5.4. Implications for evaluation

For comparable catalogue studies, the evidence supports reporting correspondence, partition agreement and taxonomy alignment as separate outcomes, with an explicit perturbation reference for stability. Reporting the full recipe and retained-input rule is necessary to interpret those outcomes. Small AMI differences alone do not establish better app discovery, categorization utility or user benefit, so the present results do not justify a deployment recommendation.

6. Limitations

The evaluation is limited to a selected Google Play catalogue without a verified probability-sampling frame; dependence among apps or developers remains unresolved. Store genres provide a flat, single-label commercial taxonomy rather than semantic ground truth, so taxonomy alignment does not establish performance on downstream user tasks. The paired design holds app identity and genre fixed while language, wording, retained text and translation condition change together, preventing a causal interpretation of the bilingual contrast. Translation quality was not independently assessed, and recorded provenance does not verify human authorship: not-flagged descriptions may include unrecorded translation. The findings therefore characterize the observed catalogue and recorded provenance strata; their transfer to broader app populations remains untested.

The experiments compare complete representation recipes that differ in tokenization, segmentation, pooling and input capacity. Native and common-content conditions also vary retained text, so their contrasts do not isolate context length; the 950-character cutoff follows an input-audit criterion rather than a performance optimum. Clustering findings are specific to the tested configurations, including the taxonomy-motivated choice of 49 clusters, and the secondary average-linkage results do not establish general algorithm rankings. The English-half control is restricted to eligible descriptions under the native lexical recipe. Its changes in content and length are unmatched to translation, limiting its relevance to neural encoders and causal translation effects. Provenance-specific refits compare strata with unequal sizes and genre compositions while retaining fixed neural representations and the global lexical map. These comparisons do not isolate provenance effects or architecture-level superiority, and small AMI differences do not establish practical user benefits.

The conditional seed intervals summarize variation across ten recorded KMeans initializations on fixed apps and representations under a small-panel approximation. They do not quantify uncertainty from app or developer sampling, embedding generation, repeated pipeline fitting or a population of possible seeds. The bootstrap multiplicity-sensitivity diagnostic identifies conditional sensitivity of historical fixed-label AMI resampling to duplicate-app weighting; it does not establish population bias or interval coverage and supplies no corrected estimator or confidence interval. Full-cohort point estimates, conditional seed intervals and diagnostic quantities are reported separately, with app-composition confidence-interval calibration remaining unresolved.

7. Conclusion

In the paired English-source and Google-translated Vietnamese catalogue, positive same-app correspondence coexists with lower cross-language partition agreement than initialization-only within-language agreement. The direction of the taxonomy contrast varies by recipe. In the larger mixed-provenance Vietnamese benchmark and its provenance-specific refits, E5, BGE and Qwen occupy the highest three observed mean positions, with condition-dependent ordering.

The native lexical English-half control shows that a larger partition deficit can occur under same-language content change, limiting translation-specific interpretations. The multiplicity diagnostic qualifies historical AMI app-resampling evidence without supplying corrected intervals. The study provides a

paired catalogue evaluation in which correspondence, partition agreement and taxonomy alignment have distinct empirical interpretations under explicit recipe and uncertainty conditions.

Data and code availability

A local reproducibility bundle has been prepared with metadata, scripts, protocols and hashes. Full description text, vector arrays and vocabulary artifacts are excluded from that bundle. Public release permissions and a public repository URL remain pending; the manuscript does not claim that data or code have been publicly released.

AI assistance disclosure

AI assistants supported code, evidence organization and manuscript drafting, including Codex assistance for this draft. These tools are not authors, and their assistance does not constitute independent human review. The human authors retain responsibility for verifying the evidence, references, interpretation and final text. The final disclosure wording and any venue-specific policy requirements require author confirmation.

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Appendix A. Representation recipes and checkpoint specifications

Table A1. Representation recipes. Mean pooling uses the attention mask. Neural tokenizers use their corresponding pinned checkpoint revisions in Table A2. PyVi segmentation is applied before tokenization where indicated. All saved vectors are L2-normalized float32.

Recipe	Pooling / representation	PyVi	Fast tokenizer	Runtime prefix	Native cap	Dimensions
TF-IDF–LSA	Truncated SVD, then L2 normalization	No	Regex word tokenizer	None	Unbounded	300
PhoBERT-v2	Attention-mask mean	Yes	No	None	256	768
ViDeBERTa	Attention-mask mean	Yes	Yes	None	512	768
Sup-SimCSE	Pooler output	Yes	No	None	256	768
BKAI bi-encoder	Attention-mask mean	Yes	No	None	256	768
E5-large	Attention-mask mean	No	Yes	Query prefix, including trailing space	512	1,024
BGE-M3	Dense CLS representation	No	Yes	None	8,192	1,024
Qwen3-Embedding-0.6B	Last-token pooling	No	Yes	None; no instruction	32,768	1,024

Table A2. Checkpoint and tokenizer revisions. Neural tokenizers use AutoTokenizer from the same checkpoint revision; tokenizer mode and segmentation are given in Table A1.

Recipe	Checkpoint / implementation	Revision
TF-IDF–LSA	local:scikit-learn/TfidfVectorizer+ TruncatedSVD	`paac-vn-tfidf-v1`
PhoBERT-v2	vinai/phobert-base-v2	`86cd7fd4c148980922ac11a2cf5e257f2ba639e1`
ViDeBERTa	Fsoft-AIC/videberta-base	`a27c8175098cc2b2b55585763fe26c60b117804c`
Sup-SimCSE	VoVanPhuc/ sup-SimCSE-VietNamese-phobert-base	`608779b86741a8acd8c8d38132974ff04086b138`
BKAI bi-encoder	bkai-foundation-models/vietnamese-bi-encoder	`84f9d9ada0d1a3c37557398b9ae9fcedcdf40be0`
E5-large	intfloat/multilingual-e5-large	`3d7cfbdacd47fdda877c5cd8a79fbcc4f2a574f3`
BGE-M3	BAAI/bge-m3	`5617a9f61b028005a4858fdac845db406aefb181`
Qwen3-Embedding-0.6B	Qwen/Qwen3-Embedding-0.6B	`97b0c614be4d77ee51c0cef4e5f07c00f9eb65b3`

The Vietnamese input is the cleaned description field (description_clean), and paired English input is the cleaned original field (description_original_clean). The lexical tokenizer pattern is `(?u)\b\w\w+\b`; the fitting settings are given in Section 3.2. Tables A1 and A2 distinguish pooling methods, preprocessing, input caps and checkpoint revisions.

Appendix B. Dataset and input audit

Table B1. Cleaned Unicode character distributions. These stored input summaries describe the mixed Vietnamese corpus and the two sides of the paired cohort, before recipe-specific input handling.

Corpus / language	N	Mean	Median	25th percentile	75th percentile	90th percentile	95th percentile	99th percentile	Maximum
Mixed Vietnamese	23883	2038.62	1874.00	1082.00	2919.50	3789.00	4104.00	4615.36	13534.00
Paired English	13657	1835.77	1670.00	983.00	2640.00	3478.00	3765.20	3938.44	7250.00
Paired Vietnamese	13657	2078.19	1888.00	1107.00	2980.00	3940.00	4266.20	4625.44	9212.00

The paired Vietnamese-to-English character ratio has mean 1.129739 and median 1.129045; the fraction of pairs with longer Vietnamese text is 0.971663.

The truncation columns in Table B2 count token-cap overflow after any character-prefix selection and preprocessing. Zero token-cap truncation does not imply that the common-content prefix retained the entire description.

Table B2. Stored token counts and truncation in all reported input conditions. Counts are measured after applicable character-prefix selection, segmentation and runtime prefix, before the final cap. Neural counts include special tokens. Lexical counts use regex word tokens. Unbounded lexical native input is not truncated. The two benchmark families use different shared-input rules, identified in the condition column.

Corpus	Recipe	Language	Condition	N	Cap	Mean tokens	Median tokens	95th percentile	Truncated N	Truncated %
Mixed Vietnamese	TF-IDF–LSA	VI	950-character common-content	23883	Unbounded	185.282	202.000	214.000	0	0.000
Mixed Vietnamese	TF-IDF–LSA	VI	Native	23883	Unbounded	436.417	402.000	883.000	0	0.000
Paired EN–VI	TF-IDF–LSA	EN	512-token	13657	512	285.039	259.000	583.000	1728	12.653
Paired EN–VI	TF-IDF–LSA	EN	Native	13657	Unbounded	285.039	259.000	583.000	0	0.000
Paired EN–VI	TF-IDF–LSA	VI	512-token	13657	512	445.519	405.000	917.000	5072	37.138
Paired EN–VI	TF-IDF–LSA	VI	Native	13657	Unbounded	445.519	405.000	917.000	0	0.000
Mixed Vietnamese	PhoBERT-v2	VI	950-character common-content	23883	256	182.411	192.000	236.000	383	1.604
Mixed Vietnamese	PhoBERT-v2	VI	Native	23883	256	430.759	393.000	879.000	16755	70.155
Mixed Vietnamese	ViDeBERTa	VI	950-character common-content	23883	512	212.570	223.000	273.000	17	0.071
Mixed Vietnamese	ViDeBERTa	VI	Native	23883	512	503.413	460.000	1019.000	10555	44.195
Mixed Vietnamese	Sup-SimCSE	VI	950-character common-content	23883	256	182.411	192.000	236.000	383	1.604
Mixed Vietnamese	Sup-SimCSE	VI	Native	23883	256	430.759	393.000	879.000	16755	70.155
Mixed Vietnamese	BKAI bi-encoder	VI	950-character common-content	23883	256	182.411	192.000	236.000	383	1.604
Mixed Vietnamese	BKAI bi-encoder	VI	Native	23883	256	430.759	393.000	879.000	16755	70.155
Mixed Vietnamese	E5-large	VI	950-character common-content	23883	512	236.117	252.000	288.000	0	0.000
Mixed Vietnamese	E5-large	VI	Native	23883	512	552.402	507.000	1113.000	11814	49.466
Paired EN–VI	E5-large	EN	512-token	13657	512	441.507	401.000	894.000	5026	36.802
Paired EN–VI	E5-large	EN	Native	13657	512	441.507	401.000	894.000	5026	36.802
Paired EN–VI	E5-large	VI	512-token	13657	512	560.628	508.000	1146.000	6770	49.572
Paired EN–VI	E5-large	VI	Native	13657	512	560.628	508.000	1146.000	6770	49.572
Mixed Vietnamese	BGE-M3	VI	950-character common-content	23883	8192	233.117	249.000	285.000	0	0.000
Mixed Vietnamese	BGE-M3	VI	Native	23883	8192	549.402	504.000	1110.000	0	0.000
Paired EN–VI	BGE-M3	EN	512-token	13657	512	438.507	398.000	891.000	4974	36.421
Paired EN–VI	BGE-M3	EN	Native	13657	8192	438.507	398.000	891.000	0	0.000

Continued on next page

Table B2 (continued)

Corpus	Recipe	Language	Condition	N	Cap	Mean tokens	Median tokens	95th percentile	Truncated N	Truncated %
Paired EN–VI	BGE-M3	VI	512-token	13657	512	557.628	505.000	1143.000	6734	49.308
Paired EN–VI	BGE-M3	VI	Native	13657	8192	557.628	505.000	1143.000	0	0.000
Mixed Vietnamese	Qwen3-Embedding-0.6B	VI	950-character common-content	23883	32768	251.978	268.000	313.000	0	0.000
Mixed Vietnamese	Qwen3-Embedding-0.6B	VI	Native	23883	32768	596.701	549.000	1206.000	0	0.000
Paired EN–VI	Qwen3-Embedding-0.6B	EN	512-token	13657	512	390.593	354.000	797.200	4075	29.838
Paired EN–VI	Qwen3-Embedding-0.6B	EN	Native	13657	32768	390.593	354.000	797.200	0	0.000
Paired EN–VI	Qwen3-Embedding-0.6B	VI	512-token	13657	512	603.913	549.000	1240.000	7321	53.606
Paired EN–VI	Qwen3-Embedding-0.6B	VI	Native	13657	32768	603.913	549.000	1240.000	0	0.000

Table B3. Genre support by experimental cohort. Companion to Main Table 1. The minimum and maximum are apps per store genre category. “Not reported” denotes an unavailable extremum, not an empty genre.

Analysis	Apps	Minimum apps per genre	Maximum apps per genre
Paired EN–VI study	13,657	113	736
Mixed-provenance Vietnamese benchmark	23,883	195	1,217
Same-language changed-text control	13,230	Not reported	Not reported
Translation-flagged refits	16,394	122	Not reported
Not-flagged refits	7,466	18	Not reported

Appendix C. Detailed experimental results

Table C1. Correspondence ranges and within-language partition agreement. Each Recipe + Input row corresponds to the same condition in Main Table 2 on 13,657 paired apps. The historical scalar 95% percentile range belongs to mean Δcos, not AMI or S; the multiplicity diagnostic does not establish its coverage. Within EN and Within VI are the mean AMIs over 45 distinct seed-pair comparisons in each language. These are the two within-language terms used in S as defined in Section 3.4.

Recipe	Input	Historical scalar range for Δcos	Within EN	Within VI
BGE-M3	Native	[0.456617, 0.458203]	0.775966	0.785221
BGE-M3	512-token	[0.456817, 0.458439]	0.791618	0.784046
E5-large	Native	[0.109762, 0.110314]	0.761508	0.767397
E5-large	512-token	[0.109762, 0.110314]	0.761476	0.767397
Qwen3-Embedding-0.6B	Native	[0.601752, 0.604107]	0.795189	0.807581
Qwen3-Embedding-0.6B	512-token	[0.598441, 0.600834]	0.796809	0.810814
TF-IDF–LSA	Native	[0.063323, 0.065561]	0.808621	0.802828
TF-IDF–LSA	512-token	[0.060524, 0.062711]	0.814841	0.810931

Table C2. Conditional seed intervals for cross-language taxonomy contrasts. Each Recipe + Input row corresponds to Main Table 3. The complete interval applies to EN minus VI taxonomy AMI, not to either individual mean. These are the existing marginal, unadjusted paired-seed t 95% intervals from ten matched-seed differences, with nine degrees of freedom, conditional on fixed apps and representations. They describe initialization variation, not app or population uncertainty.

Recipe	Input	Conditional seed interval for EN − VI
BGE-M3	Native	[0.001685, 0.008014]
E5-large	Native	[-0.022724, -0.015022]
Qwen3-Embedding-0.6B	Native	[-0.002413, 0.009078]
TF-IDF–LSA	Native	[-0.021211, -0.010161]
BGE-M3	512-token	[0.002879, 0.012741]
E5-large	512-token	[-0.022245, -0.012686]
Qwen3-Embedding-0.6B	512-token	[-0.001220, 0.010284]
TF-IDF–LSA	512-token	[-0.025046, -0.013658]

Table C3. Conditional seed intervals for Vietnamese input contrasts. Each Recipe row corresponds to Main Table 5. The complete interval applies to native minus 950-character common-content taxonomy AMI. These are the existing marginal, unadjusted matched-seed t 95% intervals from ten seed differences, with nine degrees of freedom, conditional on fixed apps and representations. They describe initialization variation, not app or population uncertainty.

Recipe	Conditional seed interval for native − common-content
BGE-M3	[-0.002830, 0.010237]
BKAI bi-encoder	[0.007316, 0.017821]
E5-large	[-0.014677, -0.005054]
PhoBERT-v2	[-0.004199, 0.005390]
Qwen3-Embedding-0.6B	[0.004932, 0.013908]
Sup-SimCSE	[0.002417, 0.008384]
TF-IDF–LSA	[0.007987, 0.015708]
ViDeBERTa	[0.005179, 0.006010]

Table C4. Provenance-specific clustering refits. Companion to Section 4.5. Translation-flagged N = 16,394; not-flagged N = 7,466. The 23 unknown-provenance apps are excluded from these refits. Means and sample seed SDs summarize ten independently initialized KMeans fits per cell on fixed representations. Ranks are descriptive within each stratum and input condition; seed SD is not an app-sampling standard error. Not-flagged does not verify human authorship.

Recipe	Input	Flagged AMI	Flagged SD	Flagged rank	Not-flagged AMI	Not-flagged SD	Not-flagged rank
TF-IDF–LSA	950-character common-content	0.479366	0.004859	4	0.516628	0.005637	4
TF-IDF–LSA	Native	0.494098	0.003694	4	0.537712	0.004977	4
PhoBERT-v2	950-character common-content	0.394383	0.004003	6	0.444310	0.005983	6
PhoBERT-v2	Native	0.396225	0.003414	6	0.446709	0.005314	7
ViDeBERTa	950-character common-content	0.021397	0.000527	8	0.034846	0.001330	8
ViDeBERTa	Native	0.026788	0.000832	8	0.045665	0.001069	8
Sup-SimCSE	950-character common-content	0.453975	0.004539	5	0.508967	0.003512	5
Sup-SimCSE	Native	0.460203	0.004272	5	0.509779	0.003888	5
BKAI bi-encoder	950-character common-content	0.353161	0.005463	7	0.432820	0.005078	7
BKAI bi-encoder	Native	0.371305	0.009177	7	0.448934	0.004807	6
E5-large	950-character common-content	0.527873	0.003254	1	0.559361	0.004549	1
E5-large	Native	0.520593	0.004627	3	0.556867	0.007364	1
BGE-M3	950-character common-content	0.523293	0.005595	2	0.549368	0.006220	2
BGE-M3	Native	0.526832	0.002519	2	0.556296	0.003790	2
Qwen3-Embedding-0.6B	950-character common-content	0.519052	0.003842	3	0.545130	0.003264	3
Qwen3-Embedding-0.6B	Native	0.527655	0.003822	1	0.555295	0.004147	3

Appendix D. Statistical diagnostics

The diagnostic retains the original 200 paired draw identities and fixed partitions. The WR series comes from stored replacement-resampling outputs; UNIQUE counts each selected app once on the corresponding support. The recorded preflight passed 280 of 280 comparisons, with maximum error $2.220446049250313 \times 10^{-16}$ against tolerance 10^{-8}. The completed diagnostic involved 377,200 AMI evaluations. These execution checks concern agreement of the implemented quantities, not inferential calibration.

The paired diagnostic centers are stable across the two halves of the stored draw sequence. For native BGE, D_{MULT} is 0.001264531, its paired median is 0.001225176, and its Monte Carlo standard error is 4.464×10^{-5}; the split-half means are 0.001298016 and 0.001231046. For the 512-token lexical condition, the corresponding mean is 0.003401710, median 0.003399396 and Monte Carlo standard error 9.024×10^{-5}, with split-half means of 0.003415053 and 0.003388368. Monte Carlo precision concerns this diagnostic's simulation average, not uncertainty from sampling apps.

App multiplicities change the empirical contingency-cell weights used by AMI. The paired comparison measures sensitivity to those weights on the selected support, conditional on fixed partitions. Individual taxonomy AMIs also have positive multiplicity differences, whereas paired taxonomy contrasts have smaller, mixed differences that can cancel. Matching repeated-label cross- and within-language component series are unavailable, so a paired component decomposition is not supported.

Table D1. Centering relative to the full-cohort S point. d_{WR} and d_{UNIQUE} are diagnostic means minus T_N. D_{MULT} is the mean paired WR-minus-UNIQUE difference. Values are diagnostic only; no column is a corrected estimate or confidence interval.

Recipe	Input	T_N	d_{WR}	d_{UNIQUE}	D_{MULT}
BGE-M3	Native	-0.053488773	0.001604922	0.000340390	0.001264531
BGE-M3	512-token	-0.064320039	0.001575563	0.000157999	0.001417564
E5-large	Native	-0.091736830	0.002085942	0.000134478	0.001951464
E5-large	512-token	-0.091655222	0.002001722	5.950e-05	0.001942223
Qwen3-Embedding-0.6B	Native	-0.092792916	0.002068456	6.482e-05	0.002003639
Qwen3-Embedding-0.6B	512-token	-0.099476153	0.002319424	6.327e-05	0.002256150
TF-IDF–LSA	Native	-0.131110868	0.002549683	-0.000250747	0.002800431
TF-IDF–LSA	512-token	-0.155594187	0.003118461	-0.000283250	0.003401710

Table D2. Diagnostic medians and Monte Carlo precision. Median offsets are the WR and UNIQUE medians minus T_N. The paired median summarizes WR minus UNIQUE within each draw. MCSE is the standard error of the simulated paired mean and is not an app-sampling standard error.

Recipe	Input	WR median $- T_N$	UNIQUE median $- T_N$	Paired median	Paired SD	Paired-mean MCSE
BGE-M3	Native	0.001758924	0.000375066	0.001225176	0.000631251	4.464e-05
BGE-M3	512-token	0.001734160	0.000187336	0.001459851	0.000761694	5.386e-05
E5-large	Native	0.001968309	0.000144271	0.001936316	0.000836905	5.918e-05
E5-large	512-token	0.001866988	1.815e-05	0.001924963	0.000830070	5.869e-05
Qwen3-Embedding-0.6B	Native	0.002064053	0.000126893	0.002051890	0.001033987	7.311e-05
Qwen3-Embedding-0.6B	512-token	0.002249693	-1.479e-05	0.002217499	0.001043871	7.381e-05
TF-IDF–LSA	Native	0.002567051	-0.000246595	0.002801472	0.001110341	7.851e-05
TF-IDF–LSA	512-token	0.003231520	-0.000354277	0.003399396	0.001276146	9.024e-05

Table D3. Stability of paired diagnostic centers across draw halves. Each column is the mean WR-minus-UNIQUE difference in the named half of the stored draw sequence. The comparison assesses simulation stability, not population coverage.

Recipe	Input	First-half mean	Second-half mean
BGE-M3	Native	0.001298016	0.001231046
BGE-M3	512-token	0.001467129	0.001367999
E5-large	Native	0.001980946	0.001921981
E5-large	512-token	0.001982609	0.001901836
Qwen3-Embedding-0.6B	Native	0.002011919	0.001995358
Qwen3-Embedding-0.6B	512-token	0.002270064	0.002242236
TF-IDF–LSA	Native	0.002798756	0.002802105
TF-IDF–LSA	512-token	0.003415053	0.003388368